SQL ASSIGNMENT 12

Subject: Database Management Systems (SQL)

Topic: Window Functions (ROW_NUMBER, RANK, DENSE_RANK, PARTITION BY)

Status: Completed

QUESTION 1

Create a table named `Playlists` (id, user_id, playlist_name, total_likes) and insert at least 8 sample rows with shared user IDs.

```
CREATE TABLE Playlists ( id INT PRIMARY KEY, user_id INT, playlist_name VARCHAR(100), total_likes INT ); INSERT INTO Playlists (id, user_id, playlist_name, total_likes) VALUES (1, 101, 'Indie Chill', 1250), (2, 101, 'Workout EDM', 3400), (3, 101, 'Acoustic Covers', 1250), (4, 102, 'Lo-Fi Beats', 5200), (5, 102, '90s Bollywood Hits', 4100), (6, 103, 'Rock Classics', 890), (7, 103, 'Jazz Vibes', 2300), (8, 103, 'Top 50 Pop', 2300);
```

QUESTION 2

Write a SQL query using `ROW_NUMBER()` and `OVER()` to assign a unique row number to each playlist ordered by `total_likes` descending.

```
SELECT playlist_name, user_id, total_likes, ROW_NUMBER() OVER (ORDER BY total_likes DESC) AS row_num FROM Playlists;
```

QUESTION 3

Use `RANK()` with `OVER()` to rank all playlists globally by `total_likes`.

```
SELECT playlist_name, user_id, total_likes, RANK() OVER (ORDER BY total_likes DESC) AS global_rank FROM Playlists;
```

QUESTION 4

Use `DENSE_RANK()` and `PARTITION BY user_id` to rank each user's playlists individually by popularity.

```
SELECT user_id, playlist_name, total_likes, DENSE_RANK() OVER ( PARTITION BY user_id ORDER BY total_likes DESC ) AS user_playlist_rank FROM Playlists;
```

QUESTION 5

Write a query using a CTE and window function to select the top 2 playlists per user based on `total_likes` (Spotify 'Your Top Playlists' feature).

```
WITH RankedUserPlaylists AS ( SELECT user_id, playlist_name, total_likes, ROW_NUMBER() OVER ( PARTITION BY user_id ORDER BY total_likes DESC ) AS rank_in_user FROM Playlists ) SELECT user_id, playlist_name, total_likes FROM RankedUserPlaylists WHERE rank_in_user <= 2;
```